

Red-shift

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 3 starred (★) questions first.

	QUESTION	ANSWER	✓
1 ★	What is red-shift?	The increase in the observed wavelength of light from galaxies that are moving away.	☐
2 ★	Explain why light from distant galaxies is red-shifted.	The galaxies are moving away from us.	☐
3	Describe how the red-shift of a galaxy depends on its speed.	The faster it is moving away, the greater the red-shift.	☐
4	Describe how the speed of a galaxy depends on its distance from us.	The further away it is, the faster it is moving away.	☐
5	Describe the Big Bang theory.	The Universe began from a very small, hot, dense region and has been expanding ever since.	☐
6 ★	Explain why red-shift is evidence for the Big Bang theory.	The galaxies are all moving apart, so they must once have been close together.	☐
7	Name two things about the Universe that scientists still do not understand.	Dark energy and dark matter	☐
8	State what observations of supernovae since 1998 suggest about the Universe.	Its expansion is accelerating.	☐
9	State what happens to the observed wavelength of light from a galaxy moving away from us.	It increases.	☐
10	State what the red-shift of distant galaxies provides evidence for.	That the Universe is expanding (the Big Bang theory).	☐